



Effects of N₂ and NH₃ remote plasma nitridation on the structural and electrical characteristics of the HfO₂ gate dielectrics

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ABSTRACT

The remote plasma nitridation (RPN) of an HfO₂ film using N₂ and NH₃ has been investigated comparatively. X-ray photoelectron spectroscopy and Auger electron spectroscopy analyses after post-deposition annealing (PDA) at 700 °C show that a large amount of nitrogen is present in the bulk film as well as in the interfacial layer for the HfO₂ film nitrided with NH₃-RPN. It is also shown that the interfacial layer formed during RPN and PDA is a nitrogen-rich Hf-silicate. The C–V characteristics of an HfO_xN_y gate dielectric nitrided with NH₃-RPN have a smaller equivalent oxide thickness than that nitrided with N₂-RPN in spite of its thicker interfacial layer.

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1. Introduction

HfO₂ is considered one of the most promising high-*k* gate dielectrics in advanced complementary metal–oxide–semiconductor (MOS) devices because of its high dielectric constant, relatively large energy band gap, and thermal stability in contact with Si [1]. However, some problems remain to be solved, such as the formation of an interfacial layer that has a low dielectric constant, the low crystallization temperature, and the threshold voltage shift caused by the fixed charge [2]. It is known that the incorporation of nitrogen into HfO₂ films is beneficial in raising its crystallization temperature, inhibiting the formation of the interfacial oxide, and improving the electrical performance of the device [3,4]. The nitridation of HfO₂ thin films can be implemented by several methods including chemical vapor deposition (CVD) of HfO_xN_y [5,6], reoxidation of an HfN film deposited by physical vapor deposition (PVD) [7], thermal annealing using NH₃ [8], and others. However, the Hf–N bonds were not stable during post-deposition annealing (PDA), especially those formed at a low processing temperature, such as reactive-sputtering of HfO_xN_y and plasma-enhanced chemical vapor deposition (PECVD) of HfO_xN_y, so that the nitrogen concentration was significantly reduced and the Hf–N bonds disappeared after annealing at 650 °C [9,10].

In this study, we have investigated HfO_xN_y gate dielectrics formed by remote plasma nitridation (RPN) of HfO₂ thin films deposited by atomic layer deposition (ALD). Plasma nitridation has been reported as an efficient method for incorporation of nitrogen into HfO₂ films at a low processing temperature [11]. Furthermore, the RPN process can minimize the plasma-induced damages by separating the reactive species from the glow zone of the plasma generation. In our experiments, remote plasmas of N₂ and NH₃ were compared as nitridation sources, and the effects on the structural and electrical characteristics of the resultant HfO_xN_y gate dielectrics were discussed.

2. Experimental

P-type Si(100) substrates with a resistivity of 4–9 Ω cm were pre-cleaned by the standard Radio Corporation of America (RCA) method, immersed in dilute HF (1%) solution for 1 min, and rinsed in de-ionized water. Films of 4.5 nm thick HfO₂ were subsequently deposited by ALD using tetrakis(ethylmethylamino)hafnium (TEMA-Hf) and ozone (O₃) at a substrate temperature of 400 °C. The nitridation of the HfO₂ films was conducted by N₂ and NH₃ remote plasma treatment for 90 s at a temperature of 500 °C using an electron cyclotron resonance (ECR) plasma system. The ECR plasma was operated at 2.45 GHz with 2 kW of microwave input power. The pressure in the chamber during the nitridation was 3 Torr with N₂ or NH₃ flow rates of 3 slm. Non-nitrided HfO₂ film was also prepared for comparison. All of the samples were subjected to post-deposition annealing (PDA) in an N₂ ambient at 700 °C for 60 s. After the PDA, X-ray photoelectron spectroscopy (XPS) anal-

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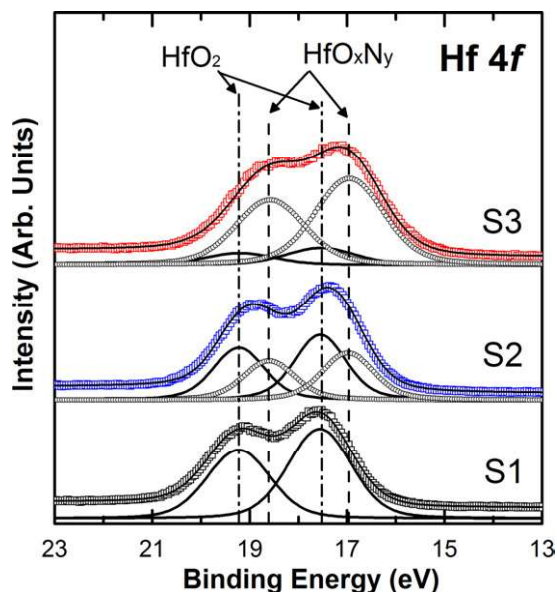


Fig. 1. XPS spectra of Hf 4f for ~4.5 nm thick HfO₂ films (S1: non-nitrided; S2: nitrided with N₂-RPN; S3: nitrided with NH₃-RPN). The PDA was performed at 700 °C for 60 s in an N₂ ambient.

ysis with a monochromatic Al K α X-ray source was carried out to study the chemical bonding states of the nitrided HfO₂ films. Auger electron spectroscopy (AES) analysis and cross-sectional transmission electron microscopy (TEM) were performed to investigate the depth profile of the nitrogen and the film thicknesses, respectively. MOS capacitors applying the HfO_xN_y gate dielectrics were also prepared with a poly-Si gate. After gate patterning, phosphorus was implanted with an energy of 50 keV and a fluence of 6×10^{15} ions/cm². Dopant activation by rapid thermal annealing (RTA) was performed at 1000 °C for 10 s in an N₂ ambient. After a metallization process and a forming gas (H₂/N₂) annealing at 400 °C for 30 min, the capacitance–voltage (C–V) characteristics were measured by an HP4275A LCR meter and the leakage current density were measured by an HP4156A semiconductor parameter analyzer.

3. Results and discussion

Figs. 1–3 show the comparative XPS spectra of the Hf 4f, N 1s, and Si 2p, respectively, for the HfO₂ films non-nitrided (S1) and those nitrided with N₂ (S2) and NH₃ (S3) remote plasma after the 700 °C PDA. The shift of the whole spectrum was calibrated by the Si 2p peak at 99.3 eV. As shown in Fig. 1, the binding energies corresponding to Hf 4f_{7/2} and Hf 4f_{5/2} for sample S1 were observed at ~17.6 and ~19.3 eV, respectively, which agree well with the reported values for HfO₂ films [8,12]. A shift to lower binding energies and a broadening of Hf 4f spectral peaks are observed for the nitrided samples (S2 and S3). It has been reported that the Hf 4f peaks for the HfO_xN_y films shift to lower binding energies as compared to HfO₂ films, indicating the presence of Hf–N bonds in addition to Hf–O bonds [6–8]. The deconvolution of the Hf 4f peaks suggests that an additional phase attributed to the HfO_xN_y film appears for S2 and S3. The Hf 4f_{7/2} peak with a binding energy of 17.0 eV corresponds to the Hf–O–N bond [12]. A comparison of relative intensities of HfO_xN_y peaks indicates that a larger amount of nitrogen is incorporated into the HfO₂ film by NH₃-RPN than by N₂-RPN and the incorporated nitrogen still exists after 700 °C PDA. However, more aggressive nitridation with NH₃-RPN by increasing the nitridation time (ex. 120 s) introduces a hafnium–silicide peak at around 14.5 eV (not shown in here).

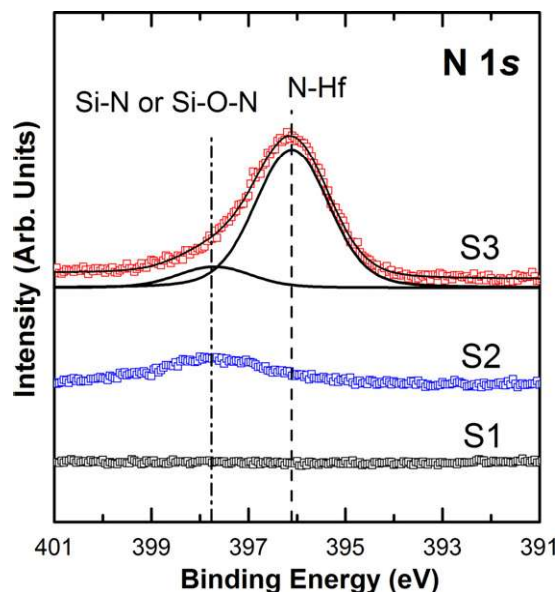


Fig. 2. XPS spectra of N 1s for ~4.5 nm thick HfO₂ films (S1: non-nitrided; S2: nitrided with N₂-RPN; S3: nitrided with NH₃-RPN). The PDA was performed at 700 °C for 60 s in an N₂ ambient.

The presence of the HfO_xN_y phase after 700 °C PDA is also confirmed by the N 1s spectral peak, as shown in Fig. 2. Sample S1 shows no peak for N 1s, whereas sample S2 shows a weak N 1s peak at 397.8 eV and sample S3 shows a noticeable peak at 396.2 eV and a weak peak at 397.8 eV. These results imply that a small amount of nitrogen exists with the bonding state of Si–N [10,11] or Si–O–N [12] for S2, whereas a large amount of nitrogen remains in the bulk film and interfacial layer for S3 with the bonding state of Hf–N [10,13] or Hf–O–N [8,12], even after 700 °C PDA.

Fig. 3 shows the Si 2p spectrum, which consists of two peaks: a strong peak at 99.3 eV that corresponds to the bulk Si 2p peak in the substrate, and a weak peak at higher binding energy, which is related to the chemical states of oxidation or nitridation. Kirsch et al. [14] reported that the feature at 102.8 eV corresponds to the Si–O

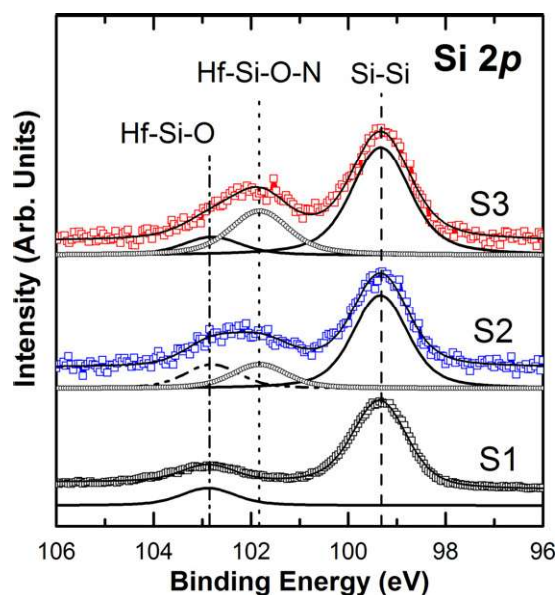


Fig. 3. XPS spectra of Si 2p for ~4.5 nm thick HfO₂ films (S1: non-nitrided; S2: nitrided with N₂-RPN; S3: nitrided with NH₃-RPN). The PDA was performed at 700 °C for 60 s in an N₂ ambient.

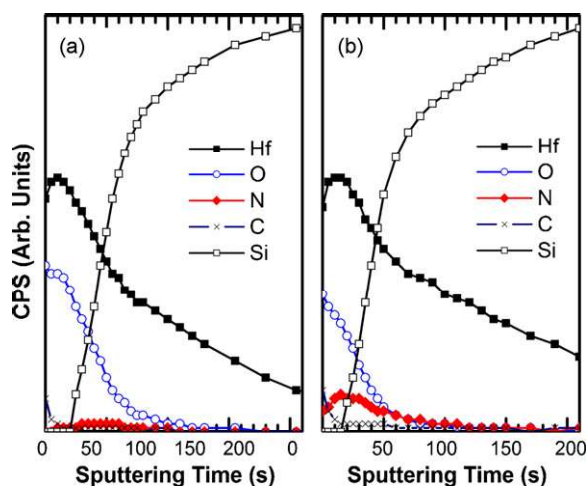


Fig. 4. AES depth profiles of the ~ 4.5 nm thick HfO_2 films nitrided with: (a) N_2 -RPN and (b) NH_3 -RPN after a 700°C PDA.

peak in a non-stoichiometric HfSi_xO_y thin film, and thus we believe that the weak peak at 102.8 eV for the sample S1 is attributed to the HfSi_xO_y interfacial layer at the HfO_2/Si interface. On the other hand, shifts in the Si $2p$ peak towards a lower binding energy are related to the nitridation of the interfacial layer for S2 and S3. The deconvolution results suggest that a new peak at ~ 101.8 eV is attributed to Hf–Si–O–N bonding state and more Hf–Si–O–N bonds are formed at interfacial layer for S3 than for S2.

Fig. 4 shows the nitrogen depth profiles of the HfO_xN_y films after PDA, which are obtained by AES analysis. The relative comparison of nitrogen contents confirms the effectiveness of NH_3 -RPN for nitrogen incorporation into an HfO_2 film. Using N_2 -RPN, a small amount of nitrogen is mainly distributed at the $\text{HfO}_x\text{N}_y/\text{Si}$ interface rather than in the bulk film, whereas a large amount of nitrogen is present in the bulk film for the NH_3 -RPN after 700°C PDA, which means that NH_3 -RPN is superior in the nitridation of HfO_2 films and in the thermal stability of resultant HfO_xN_y film.

Fig. 5 shows cross-sectional TEM images of the HfO_2 and HfO_xN_y films after the 700°C PDA. The physical thickness of the interfacial layer (IL) for the non-nitrided HfO_2 film is about 1.2 nm, and it is reduced to 1.0 nm for N_2 -RPN. It is believed that the nitrogen incorporation during the N_2 -RPN suppresses the oxygen diffusion through the HfO_2 film and inhibits an interfacial reaction with the Si substrate during the PDA. However, the HfO_2 film nitrided with NH_3 -RPN shows an increased interfacial layer thickness of ~ 1.9 nm

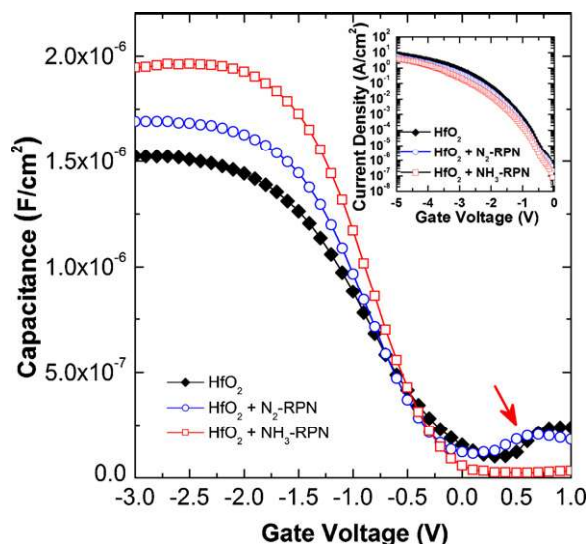


Fig. 6. Capacitance–voltage (C – V) and current–voltage (I – V) characteristics of MOS capacitors having the HfO_2 and HfO_xN_y gate dielectrics. The gate dielectrics underwent PDA at 700°C for 60 s and dopant activation annealing at 1000°C for 10 s in an N_2 ambient.

and a reduced thickness of the HfO_xN_y film. We infer that dissociated species, such as $\text{NH}_2^{+/-}$ and NH^0 , in the HfO_2 film from the NH_3 remote plasma do not suppress the oxygen diffusion but enhance the growth of Hf–silicate or $\text{HfSi}_x\text{O}_y\text{N}_z$ at the $\text{HfO}_x\text{N}_y/\text{Si}$ interface due to the additional thermal budget during the NH_3 -RPN, which is similar to the thermal annealing of the HfO_2 film using NH_3 gas [8,15].

The electrical characteristics of the MOS capacitors with HfO_2 or HfO_xN_y gate dielectrics are shown in Fig. 6. The C – V characteristics of the MOS capacitors were measured at a relatively low frequency of 10 kHz, because an anomalous shape of the C – V curve caused by the interface states and near-interface traps is more clearly shown at the low frequency [16]. The equivalent oxide thickness (EOT) was extracted by fitting the C – V curves in Fig. 6 with the simulated C – V curve that accounts for quantum mechanical effects. Although exact evaluation of the EOT is difficult due to the frequency dispersion, values of ~ 2.0 , ~ 1.75 , and ~ 1.45 nm can be extracted from the 10 kHz C – V data for the HfO_2 film and the HfO_xN_y films formed by N_2 -RPN and NH_3 -RPN, respectively. The decrease of the EOT for N_2 -RPN films from that of non-nitrided film can be explained by the reduced interfacial layer thickness and the formation of the

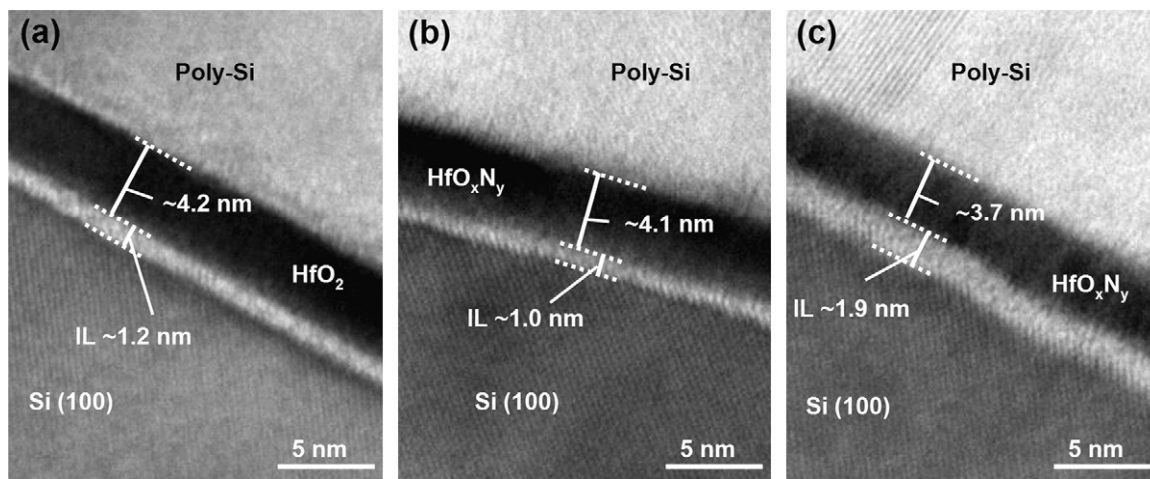


Fig. 5. Cross-sectional TEM images of the HfO_2 films: (a) non-nitrided and nitrided with (b) N_2 -RPN and (c) NH_3 -RPN after a 700°C PDA.

some portion of HfO_xN_y having a higher dielectric constant than a pure HfO_2 film. It is noteworthy that the HfO_xN_y gate dielectric formed by NH_3 -RPN shows a further reduced EOT than that formed by N_2 -RPN, in spite of its thicker interfacial layer. This result implies that the NH_3 -RPN incorporates more nitrogen into the HfO_2 film, as proved by the XPS and the AES analysis, and the incorporated nitrogen is still exist in the film with the phase of HfO_xN_y even after dopant activation annealing at 1000°C for 10 s. The reduced EOT for the NH_3 -RPN can be further explained by the thickness reduction of HfO_xN_y film, as shown in Fig. 5(c), and the chemical states of the interfacial layer. Maikap et al. [17] reported the interfacial characteristics of HfO_2 on NH_3 - and N_2O -plasma treated Si substrates; the resultant interfacial layer was a nitrogen-rich Hf-silicate with a relatively high dielectric constant (above 10). Another important feature from the C - V curves is an unexpected increase of depletion region capacitance (indicated by an arrow in Fig. 6) for the HfO_2 and the HfO_xN_y formed by N_2 -RPN. The anomalous increase (hump) of capacitance at the depletion region (or weak inversion region for low frequency C - V measurement) may be attributed to either an increase in minority carrier generation due to the diffusion of impurities from the gate dielectrics into the substrate [16], or to the interaction of a high density of acceptor-like slow interface states and near-interface oxide traps [16,18]. However, the hump of capacitance at the depletion region is not shown for the HfO_xN_y gate dielectric formed by NH_3 -RPN implying these impurity diffusion and traps are negligible. More detailed analysis on the anomalous increase of depletion capacitance and its origin will be discussed in the other paper. The inset in Fig. 6 shows that the leakage current densities of the HfO_xN_y gate dielectrics are not deteriorated but slightly decreased by N_2 - and NH_3 -RPN.

4. Conclusion

In conclusion, we have investigated the effects of N_2 and NH_3 remote plasma nitridation on the structural and electrical characteristics of the HfO_2 gate dielectrics. AES and XPS analyses show that a small amount of nitrogen is mainly distributed at the HfO_xN_y /Si interface rather than in the bulk film for the N_2 -RPN. However, a large amount of nitrogen is present in the bulk film as well as interfacial layer for the NH_3 -RPN after the 700°C PDA. The electrical characteristics of the MOS capacitors show that the HfO_xN_y gate dielectric with NH_3 -RPN has a smaller EOT than that with N_2 -RPN

even after 1000°C activation annealing in spite of its thicker interfacial layer. The results suggest that the NH_3 -RPN is superior in both the nitridation of HfO_2 films and the thermal stability of resultant HfO_xN_y film with a high dielectric constant and a low defect density.

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References

- [1] J. Robertson, Rep. Prog. Phys. 69 (2006) 327.
- [2] M. Houssa, L. Pantisano, L.A. Ragnarsson, R. Degraeve, T. Schram, G. Pourtois, S. De Gendt, G. Groeseneken, M.M. Heyns, Mater. Sci. Eng. R 51 (2006) 37.
- [3] M. Koyama, A. Kaneko, T. Ino, M. Koike, Y. Kamata, R. Iijima, Y. Kamimuta, A. Takashima, M. Suzuki, C. Hongo, S. Inumiya, M. Takayanagi, A. Nishiyama, Tech. Dig. Int. Electron Devices Meet. 849, 2002.
- [4] K.-M. Chang, B.-N. Chen, S.-M. Huang, Appl. Surf. Sci. 254 (2008) 6116.
- [5] C.H. Choi, T.S. Jeon, R. Clark, D.L. Kwong, IEEE Electron Device Lett. 24 (2003) 215.
- [6] J.-H. Park, J.-S. Hyun, B.-C. Kang, J.-H. Boo, Surf. Coat. Technol. 201 (2007) 5336.
- [7] M. Liu, L.D. Zhang, Q. Fang, J.P. Zhang, X.J. Wang, G. He, J. Phys. D: Appl. Phys. 42 (2009) 195304.
- [8] M.-H. Cho, K.B. Chung, C.N. Whang, D.-H. Ko, J.H. Lee, N.I. Lee, Appl. Phys. Lett. 88 (2006) 202902.
- [9] K.B. Chung, G. Lucovsky, W.J. Lee, M.-H. Cho, H. Jeon, Appl. Phys. Lett. 94 (2009) 042907.
- [10] J.F. Kang, H.Y. Yu, C. Ren, M.-F. Li, D.S.H. Chan, H. Hu, H.F. Lim, W.D. Wang, D. Gui, D.-L. Kwong, Appl. Phys. Lett. 84 (2004) 1588.
- [11] G.K. Dalapati, A. Sridhara, A.S.W. Wong, C.K. Chia, D.Z. Chi, Appl. Phys. Lett. 94 (2009) 073502.
- [12] T.J. Park, J.H. Kim, J.H. Jang, K.D. Na, C.S. Hwang, J.H. Yoo, J. Appl. Phys. 104 (2008) 054101.
- [13] X.J. Wang, L.D. Zhang, M. Liu, J.P. Zhang, G. He, Appl. Phys. Lett. 92 (2008) 122901.
- [14] P.D. Kirsch, C.S. Kang, J. Lozano, J.C. Lee, J.G. Ekerdt, J. Appl. Phys. 91 (2002) 4353.
- [15] J.L. Gavartin, A.L. Shluger, A.S. Foster, G.I. Bersuker, J. Appl. Phys. 97 (2005) 053704.
- [16] P.C. McIntyre, D. Chi, C.O. Chui, H. Kim, K.I. Seo, K.C. Saraswat, R. Sreenivasan, T. Sugawara, F.S. Aguirre-Testado, R.M. Wallace, ECS Trans. 3 (2006) 519.
- [17] S. Maikap, J.-H. Lee, R. Mahapatra, S. Pal, Y.S. No, W.-K. Choi, S.K. Ray, D.-Y. Kim, Solid-State Electron. 49 (2005) 524.
- [18] H.W. Chen, F.C. Chiu, C.H. Liu, S.Y. Chen, H.S. Huang, P.C. Juan, H.L. Hwang, Appl. Surf. Sci. 254 (2008) 6112.